Magic Beams

In an enchanted kingdom, the land beneath the capital is etched with an ancient rectangular grid of runes. Along these runes flow powerful magic beams, which are long streams of shimmering energy that keep the city alive. Bound by old enchantments, magic beams are one rune wide and can stretch across several runes in length. When freed from the enchantments, a beam travels forever in one direction — North, South, East, or West — without ever turning, clearing the runes it occupied. But if any rune in its path is not completely empty, a catastrophic magical surge will occur. No two beams may ever overlap.

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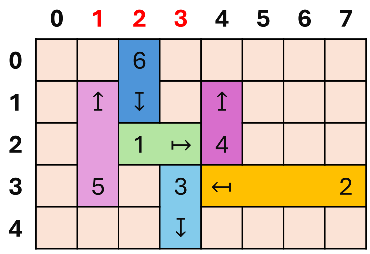
One night, the Great Portal of Safekeeping begins to flicker. The Council of Mages foresees disaster unless a stabilising corridor is opened beneath the city. To restore balance, one or more consecutive columns of runes, called the *chosen columns*, must be completely cleared of magic beams. The task falls to Eldrin, a wizard famous for his perfect understanding of cause and effect.

Eldrin descends into the rune grid and studies the glowing beams. He quickly realises the challenge: Freeing one magic beam may require freeing others first, because some beams block the path of others. Eldrin cannot bend the beams or change their directions. He may only free them, one at a time.

Can you help Eldrin? Your mission is to decide which magic beams must be freed and determine the exact order in which they should be freed, so that the chosen columns stand completely empty, allowing the Great Portal to stabilise.

Let us see an example in the rune grid depicted on the right, which has 5 rows and 8 columns. Columns 1, 2 and 3 are the chosen ones. There are 6 magic beams, denoted by $1, 2, \ldots, 6$, whose directions are specified by arrows.



- In order to clear the chosen columns, beams 1, 3, 4, 5, and 6 must be freed. Beam 2 can remain where it is.
- Beams 3, 4 and 5 can be freed at any time. Beam 4 has to be freed before beams 1 and 6. After beam 4 is freed, beam 1 can be freed. After beam 1 is freed, beam 6 can be freed.
- Knowing that when there are several beams that can be freed Eldrin always choses the beam with the lowest number, the order in which magic beams should be freed is: 3, 4, 1, 5, and 6.

Task

Write a program that, given the description of the rune grid, the chosen columns, and the information about the magic beams, indicates whether the chosen columns can be emptied of magic beams. If they can, the program determines the order in which the magic beams (that need to be freed) should be freed. Whenever there are several beams that can be freed at the same time, the beam with the lowest identifier should be freed first.

Input

On the input first line there is an integer, T, which represents the number of test cases.

For each test case there is a first line containing two integers, R and C, which denote the number of rows and the number of columns of the rune grid. Rows and columns are identified by integers, ranging from 0 to, respectively, $R-1$ and $C-1$. The cell in the upper left corner is in row 0 and column 0. The second line has also two integers, N and L, specifying that the stabilising corridor comprises N columns and L is the leftmost (westernmost) chosen column.

The third line has an integer, B, which is the number of magic beams. B lines follow, each one describing a magic beam and containing three integers, r_i, c_i, and l_i, and a letter d_i, which is `N` (North), `S` (South), `E` (East) or `W` (West), for every $i = 1, \ldots, B$. The i^{th} of these lines indicates that the magic beam, identified by i, occupies the cell in row r_i and column c_i, has length l_i, starting from (r_i, c_i), extends in the direction specified by d_i and travels in that direction.

It is guaranteed that, for the given inputs, the corridor and all magic beams are completely inside the grid.

Constraints

$1 \leq T \leq 12$	Number of test cases
$2 \leq R \leq 200$	Number of rows
$2 \leq C \leq 200$	Number of columns
$1 \leq N \leq 10$	Number of chosen columns
$1 \leq B \leq 10\,060$	Number of magic beams
$1 \leq l_i \leq 20$	Number of cells occupied by magic beam i (for every $i = 1, \ldots, B$)

Output

The output consists of T lines, one for each test case:

- If the chosen columns cannot be emptied of magic beams, the line has the word `Disaster`;
- If no beam needs to be freed, the line has the words `False alarm`;
- Otherwise, the line contains the magic beams that need to be freed, separated by one space, in the order in which they should be freed.

Sample Input 1

```
1
100 100
10 37
1
43 91 20 W
```

Sample Output 1

```
False alarm
```

Sample Input 2

```
2
5 8
3 1
6
2 2 2 E
3 7 4 W
3 3 2 S
2 4 2 N
3 1 3 N
0 2 2 S
10 10
1 4
4
5 6 2 W
6 3 3 N
3 3 2 E
2 5 3 S
```

Sample Output 2

```
3 4 1 5 6
Disaster
```